

JESSICA KUPPAN

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EDUCATION

Ph. D (Biomedical Engineering), Washington University in St. Louis (WUSTL), St. Louis, MO	Anticipated 2026
M.S (Biomedical Engineering), Washington University in St. Louis (WUSTL), St. Louis, MO	2022
B.A. in Chemistry and Concentration in Bioinformatics, CUNY Hunter College, New York, NY	2019

RESEARCH EXPERIENCE

<i>Michael Vahey Lab, Washington University in St. Louis, St. Louis, MO</i>	Fall 2019 - Present
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- Investigating epitope accessibility of complement deposition and B cell activation on respiratory viruses
- Executed a multi-year project involving cloning, PCR, cell culture, and quantitative image analysis
- Results of this work led to a publication in *eLife* and funding from an NIH R01 and NSF Career Grant

<i>Mande Holford Lab, CUNY Hunter College, New York, NY</i>	Summer 2017 – Summer 2019
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- Investigating *in vivo* delivery methods for bioactive and novel peptides in venomous marine terebrid snails with therapeutic applications. Developed viral nanocontainers for drug delivery
- Contributed to biochemistry research published in *Bioconjugate Chemistry* and presented at conferences

<i>Michael Vahey Lab, Washington University in St. Louis, St. Louis, MO</i>	Summer 2018
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- Washington University in St. Louis Summer Engineering Fellowship (WUSEF)
- Developed a droplet microfluidic device to measure the enzymatic activity of neuraminidase in Influenza A

<i>Harlem Research Lab, Urban Barcode Project (UBP), New York, NY</i>	Fall 2014 - Summer 2015
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- Conducted biodiversity assessments through DNA barcoding and bioinformatics
- Contributed to bioinformatics research published in *PLOS One* and awarded grand prize at the Cold Spring Harbor UBP competition

LEADERSHIP AND PROFESSIONAL EXPERIENCE

<i>McKelvey Engineering Graduate Ambassador Program, Washington University</i>	2025-Present
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- Representative for McKelvey Engineering at various conferences and events to support prospective, new, and current students through email, virtual events, and in-person events on campus.

<i>Immunophotonics Startup Intern, BioSTL, Saint Louis, MO</i>	2022- 2023
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- Developed pipeline to investigate future cancer therapeutic and applications
- Evaluated and tested novel cancer therapeutic in vitro and in vivo (cancer cell lines and animal models)

<i>Fellow, InSITE, Washington University in St. Louis</i>	2022-2023
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- Worked on interdisciplinary consulting teams to advise startup clients
- Developed marketing and go-to-market strategies for early-stage ventures including Travelers who Run

<i>VP of Community Service and Civic Engagement, Graduate Professional Council</i>	2022-2023
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- Represented and advocated the interest of graduate and professional students across all schools at WashU
- Spearheaded collaborative initiatives with civic organizations and university-wide programming

<i>DEI and Peer Mentorship Chair, Biomedical Engineering Doctoral Council</i>	2020 - 2023
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- Founded and led the first funded peer mentorship program for incoming BME PhD students at WashU
- Co-authored the engineering school's "Call to Action" to promote diversity, equity, and inclusion at the

<i>Peer Mentor, McKelvey First gEN program</i>	Sept. 2020 – May 2022
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- Mentored first-generation, undergraduate engineering students through academic and professional support
- Leveraged personal and leadership experience to provide students support, empowerment, and guidance throughout their transition to college

<i>Teacher, WUSTL Young Scientist Program (YSP)</i>	Sept. 2019 – May 2020
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- Provided one-on-one academic and personal mentorship to students for the school year

- Developed lesson plans aimed at preparing for college and encouraging them to pursue STEM careers

WUSEF Volunteer, WUSTL College Prep Program

June 2018

- Introduced the engineering design process and helped 25 high school students to design, build, and test their own prosthetic hand using common, everyday materials.

Maximizing Access to Research Careers (MARC) Peer Mentor

Spring 2017 – Summer 2019

- Mentored junior MARC scholars and Science 200 students at CUNY Hunter College throughout their research projects in Holford lab

PUBLICATIONS

Kuppan, J., et al. (2021), A morphological transformation in Respiratory Syncytial Virus leads to enhanced complement deposition, *eLife* 10, e70575

Kelly M.P., et al. (2020), Induced disassembly of a Virus-Like Particle under physiological conditions for venom peptide delivery. *Bioconjugate Chemistry*, 32(1), 111-120

Marizzi C, Florio A, et al. (2018), DNA barcoding Brooklyn (New York): A first assessment of biodiversity in Marine Park by citizen scientists. *PLoS ONE*, 13(7), e0199015

CONFERENCE PRESENTATIONS

Kuppan, J., Mitrovich, M, Vahey, M.D. (2025), Investigating the contributions of epitope accessibility in the activation of RSV-Specific B cells, Montreal, Canada

Kuppan, J., Vahey, M.D. (2023), Dissecting mechanisms of epitope recognition using engineered B cells. Biomedical Engineering Society, Oral Presentation, Seattle, WA

Kuppan, J., Vahey, M.D. (2023), Dissecting the contributions of epitope accessibility in B cell activation, Poster Presentation, San Diego, CA

Kuppan, J., Vahey, M.D. (2021), A morphological transformation in Respiratory Syncytial Virus leads to enhanced complement deposition, American Society for Virology, Virtual Oral Presentation

Kuppan, J., Vahey, M.D. (2018) Droplet-based Microfluidics for the Encapsulation and Detection of Neuraminidase from Influenza. Annual Biomedical Research Conference for Minority Students, Indianapolis, IN

Kuppan, J., Kelly, M.P., et. al (2018). Developing a Dissociative Viral Nanocontainer for Peptide Drug Delivery. Mount Sinai Undergraduate Research Symposium, New York, NY

HONORS AND AWARDS

ASV 2025 Graduate Student Travel Award 2025

WUSTL Olin Cup and Elevator Pitch Winner 2024

- Olin Cup: Awarded first amongst 12 teams that presented their business pitches to a group of 60 judges
- Olin Elevator Pitch: Awarded first amongst over 200 teams for a two-minute video pitch contest

WUSTL BME Chair Doctoral Fellowship 2019 – 2024

- Fellowship for graduate students who would contribute to the diversity of BME doctoral education
- Awarded an additional \$5k above the standard PhD stipend for up to five years

WUSTL Pivot 314 Fellowship 2022 – 2023

- Year long program offers graduate students curated programming focused on professional development
- Awarded \$3000 for the summer internship

WUSTL BME Doctoral Leadership Award 2023

- Award given to doctoral students who has shown school-level or university-level leadership
- Awarded a one-time payment of \$500

NIH Maximizing Access to Research Careers (MARC) U-STAR Fellow 2017 – 2019

- Program that prepares underrepresented students in biomedical science for selective PhD training
- Awarded \$12k stipend for two years

CUNY Hunter: Jenny Hunter Honors Scholar 2015 – 2019

- Honors cohort that recognizes high level academic achievement and potential
- Received \$1000 tuition award for four years

NYS Scholarship for Academic Excellence 2015 – 2019

- Provides college tuition assistance to outstanding New York State high school graduates
- Awarded \$1500 tuition award for four years

Cold Spring Harbor: UBP Grand Prize Winner 2015

- Program to introduce high school students to DNA barcoding and pursue STEM careers
- Awarded grand prize winner out of 65 high school teams across New York City